

Discovery & Design Report

Original Concept

Problem:

Barbell squatting often causes shoulder pain and bar slippage, especially in low-bar positions. Traditional foam pads are bulky and shift during use, while grip shirts are expensive, restrictive, and not versatile. Lifters want comfort and stability without sacrificing bar control or appearance.

Solution:

GripSkin is a thin, reusable film with microgel cushioning and a textured surface that enhances grip. It adheres lightly to the lifter's back or shirt, resists sweat, and remains nearly invisible during training. Unlike bulky pads, it adds stability and comfort without changing the bar's feel or lifter's form.

Customer Value:

GripSkin gives lifters a discreet, performance-based solution that combines comfort, control, and style. It's affordable, compatible with any gear, and easy to apply or remove. By solving two problems at once (pain and slippage), GripSkin makes squatting safer, cleaner, and more confident for everyday gym users.

Customer Discovery

How/Why we Talked with Potential Customers

We conducted customer discovery interviews with potential customers within our circle of acquaintances who we knew were active gym-goers and had meaningful experience with weightlifting. Our goal was to engage individuals who could provide informed perspectives on squat bar functionality, convenience, and pain points commonly experienced during workouts. We specifically sought out participants with an extensive history of weightlifting. We sought out those who not only use gym equipment regularly but also have developed personal preferences and insights about what works and what doesn't. Additionally, we focused our efforts on individuals local to the Corvallis area to ensure our early research remained practical and relevant to our target environment. Since we plan to initially market and test our product at Dixon Recreation Center, we prioritized interviewing younger gym members, as they represent our primary target segment for early adoption. By concentrating on this localized and age-specific group, we aimed to streamline product development, gather more actionable feedback, and ensure our design decisions align closely with the needs of our first intended users.

Customers Talked With

Name	Real/Virtual	Why we talked to them	Key Observations
Erika Price	Virtual	They go to the gym at least once a week, and exercise their legs fairly frequently. They also have a history of going to the gym since high school, although they have taken breaks on and off. They also have experience using different squat support devices (pads, belts, etc.)	1) Validates the importance of the adjustable thickness of the protection device. Some people want more thickness in their support than others. 2) The value assigned to the importance of making this accessible and useful to a wide variety of people. 3) Balance is a very large priority for deciding to squat. If they can't balance the weight, then they feel like they are falling backwards, and are reluctant to squat based on that.
Ryan Ngyuen	Real	Very proficient powerlifter and someone who squats very frequently. Could even be considered a professional in the field we are in. Goes to the gym 4 times a week and squats twice a week extensively using the low bar technique. Has very high opinions against using squat pads so wanted to see his input on our invention.	1) Understands the tension of having heavy weights constantly on the shoulders while squatting 2) Sees that our product is not just a flimsy squat pad 3) Appreciates the aspect of it gripping to the shirt as he has gone through many times of bars slipping off his back 4) Urges us to ensure that the low bar squat mechanic is not messed with, with the utilization of this pad

Competitor Research

How/Why we Conducted Competitor Research

We conducted competitor research to better understand the current market landscape and identify existing solutions that address the pain and discomfort associated with squatting. Our goal was to analyze competitors not only to map the range of available products but also to pinpoint gaps that our concept could fill through improved design, functionality, or comfort. We began by examining different types of squat pads ranging from basic foam designs to molded plastic and ergonomic bar supports. We wanted to assess how each pad attempted to make squatting more accessible or less painful. From there, we expanded our research to include equipment and exercises that individuals use as substitutes for squats, such as hack squat machines, Smith machines, and leg presses. This broader analysis helped us understand why many gym-goers avoid traditional squatting altogether and what features or limitations of existing alternatives influence that decision. By comparing these products and substitutes, we were able to identify opportunities to enhance both comfort and practicality while maintaining the performance benefits of a standard squat.

Types of Competitors

Type	Example	Hyperlink for Example	Strengths	Weaknesses
Machine based compound lifts	Hammer Strength Plate-Loaded Hack Squat Machine	Hammer Strength Hack Squat Machine	Keeps back straight by use of the sled, making squatting safer. — Shoulder cushions rest on the more sturdy deltoid muscle	Less Overall leg and lower back development than barbell squat — Doesn't build balance, coordination, and core strength like standard squats
Molded plastic shoulder supports	Black Mountain Products Stabilizing Squat Pad	Black Mountain Products Stabilizing Pad	Distributes Weight across more surface area — Locks the bar into one position — Ergonomic shape to mesh with back	Mold isn't adjustable, and is one size fits all — Plastic digs into shoulders making squatting uncomfortable — Very noticeable and can be embarrassing
Foam Squat	Gym Reapers	Gym	Added comfort	Messes up center of

Bar Pads, Pads with a neck divot		Reaper Squat Pad	— Pressure Relief, Divot for neck to go into — Dampens the weight of the squat bar over more surface area	gravity — Very Noticeable and can be embarrassing to use — Wears out quickly — Doesn't allow for low bar technique
Straight pads	Harbinger NeoTek Squat Pad	Harbinger NeoTek Pad	Bulky and takes load off body — Same width throughout — Foam spreads and distributes weight over larger area	Very Limited low bar squat technique — Very easily slips off back — Very loud
More Comfortable Squat Bars	Rogue Fitness Safety Squat Bar	Rogue Fitness Squat Bar	Distributes the squatting weight more to the shoulders, which are usually stronger and more developed than the neck and upper traps — Handles to better stabilize and secure the weight on your back	Immobile. The squat bar is difficult and impractical to bring around for personal use outside of a set location. — Very noticeable, and can be embarrassing to use due to the assistance it provides over regular squatting.
Dumbbell / bodyweight based leg movements	Bulgarian Split Squat	Bulgarian Split Squat Demo Video	Good for fixing muscle imbalances — Safer for spine than standard squatting	More difficult to stabilize yourself — Harder to progressively overload — Less activation of Quads than standard squatting — Uncomfortable foot positioning on bench

Other Research

Our team conducted several additional research activities to refine our business model hypotheses and expand on the in class activities we did to gauge potential customer segments, competition, and feasibility. For our first activity, we focused on developing our understanding of the material feasibility of GripSkin, exploring options like medical-grade silicone, micro-suction mesh, and skin-safe adhesives. We wanted to learn whether the design could balance comfort, flexibility, and durability without becoming bulky or expensive. We found that existing materials used in sports and medical products already meet these standards, confirming that GripSkin could be manufactured using readily available, low-cost materials while maintaining high performance and hygiene.

Next, we conducted secondary research to expand on our in-class customer segment exploration, further developing our understanding of the target market. This activity focused on how lifters, particularly younger gym-goers, perceive squat accessories. Through analysis of online fitness communities, product reviews, and social media discussions, and our own primary research efforts, we found that discretion and aesthetics are just as important as comfort. Many lifters avoid traditional squat pads because they are viewed as bulky or “beginner-like,” which can negatively impact confidence and usage rates. These findings reinforced our initial hypothesis that a subtle, minimalist solution such as GripSkin would strongly resonate with young lifters who value both performance and visual appeal.

Lastly, our competitor expansion research activity helped us broaden our view of alternative solutions. We examined how lifters substitute squats with machine-based lifts and bodyweight movements to avoid bar discomfort or instability. This revealed that GripSkin doesn't just compete with pads, it competes with entirely different exercise methods that people use to bypass pain. Observing this helped us refine our differentiation hypothesis: GripSkin fills a unique gap by making barbell squats more comfortable and accessible without changing equipment or technique. Together, these activities reinforced that the product is technically feasible, highly desirable, and strategically positioned to meet the needs of modern gym-goers.

Research Discussion

We conducted research to check our customer segments, ensuring that our idea of who our primary audience would be matched with the reality of the fitness area around weightlifting in Corvallis. Through our primary research, we gauged Corvallis gym-goers' reactions and perspectives regarding our solution. Some important learnings arose from this that forced us to reevaluate our solution hypothesis. Specifically, we found that creating an accessible and versatile product was very important to potential customers, as they found that the tools that are

common in the market are not very adjustable and don't work well with their body types. This helped to redefine our solution compared to competitors, and what unique value our product offered over others in the market.

We also performed primary research in the form of customer interviews focused on younger gym-goers, especially at Oregon State's Dixon Recreation Center. We did this to gauge potential customer desirability. Our findings validated our initial exploratory research, where we made our hypothesis that young lifters highly value discretion and may avoid bulky pads or flashy equipment that make them look inexperienced. The interviews supported our initial ideas and beliefs that GripSkin's invisible, minimalist design meets a key emotional and functional need in this demographic.

Lastly, we attempted to further our understanding of our product's uniqueness and of the unique value proposition that it could offer compared to existing solutions. Based on research indicating that many lifters avoid barbell squats due to pain or instability, we expanded our list of potential competitors to include substitutes for squatting like machine-based lifts and dumbbell/bodyweight exercises. This insight reinforced that GripSkin competes not only with other pads but with any product or method used to avoid bar discomfort. Together, these findings strengthened our confidence that GripSkin is both desirable and technically feasible within the current market.

Differentiation Hypothesis

Our first key differentiation hypothesis is that the GripSkin wrap pad's full-length, silicone construction provides a superior balance of comfort, bar stability, and even pressure distribution compared to any current squat pad or bar accessory. Traditional foam or molded plastic pads fail to provide consistent cushioning, foam pads compress unevenly and distort bar balance, while hard plastic designs dig into the shoulders and cannot adjust to different body shapes. In contrast, the GripSkin wrap pad uses medical-grade transparent silicone with two types of surfaces for each side of the wrap. On one side, a micro-suction interior mesh, which forms a vacuum-seal grip around the barbell and prevents any slipping or rolling during lifts. This ensures that the bar remains perfectly centered and stable while still delivering a soft, stable interface between the bar and the lifter's back. On the other side, the pad's 1/8-inch tactile bump texture distributes weight uniformly across the traps and upper back, reducing pressure points without adding unnecessary bulk or instability, something our potential customer interviews communicated was very desirable. Technically, this innovation is both feasible and realistic, as it leverages existing silicone molding and suction technologies already proven in athletic and medical applications. The result is a product that delivers genuine performance comfort without sacrificing the precision and balance demanded by serious lifters, a combination no current competitor provides.

Our second differentiation hypothesis centers on GripSkin's ability to deliver comfort and functionality in a discreet, customizable, and travel-ready design. Most existing solutions such as bright foam pads, rigid molded supports, or large alternative equipment like hack squat machines, are either bulky, visually obtrusive, or immobile. These products make users stand out in ways that often feel awkward or "beginner-like," discouraging consistent use despite their intended comfort benefits. The GripSkin wrap pad eliminates that problem through a transparent, minimalist form factor that maintains the professional look of a bare barbell. Its wrap-and-tie system allows users to fine-tune the tightness and feel of the pad, providing adjustable tension that accommodates individual preferences and body types. Additionally, the locking ring anchor ensures security under heavy loads, while the lightweight silicone body can be unwrapped, cleaned, and stored in seconds, fitting easily in a gym bag. This level of portability and discretion directly addresses unmet needs among everyday gym-goers and university lifters, users who want relief from bar pain but refuse to compromise on appearance, performance, or convenience. By combining comfort with subtlety and mobility, the GripSkin wrap pad redefines what a squat pad can be: not a crutch, but a seamless enhancement for serious training.

Product Concept Summary

The GripSkin wrap pad is a wrap-style squat bar attachment engineered for lifters who want comfort, control, and discretion without the bulk of traditional foam pads. The pad spans a total of 45 inches, spanning more than the entire contact zone of a standard squat pad (~17–18 inches) and is made from transparent, medical-grade silicone. It wraps seamlessly around the bar's surface, providing uniform cushioning and grip across the upper-back contact area. At the left end, the pad features a reinforced metallic locking ring that anchors it securely to the barbell. From this point, the user wraps the silicone layer tightly around the bar, overlapping it across the bar's curvature. The right end of the wrap is designed to tie back on itself, creating a secure, self-fastened knot that holds firm under compression.

The wrap maintains an approximate total diameter of 1¾" when applied around singly wrapped through the length of the squat bar's contact zone, with an inner micro-suction mesh that conforms to the bar's surface. This design creates a natural vacuum-like seal as the wrap tightens, locking the bar in place without adhesives or clamps. The outer layer features ⅛-inch tactile bumps that come into contact with the lifter's upper back and traps. These micro-textures enhance grip on the neck, prevent slipping, and distribute bar pressure evenly. Despite its technical precision, the pad remains ultra-thin (~¾"), clear, and minimalist, almost blending invisibly with the bar.

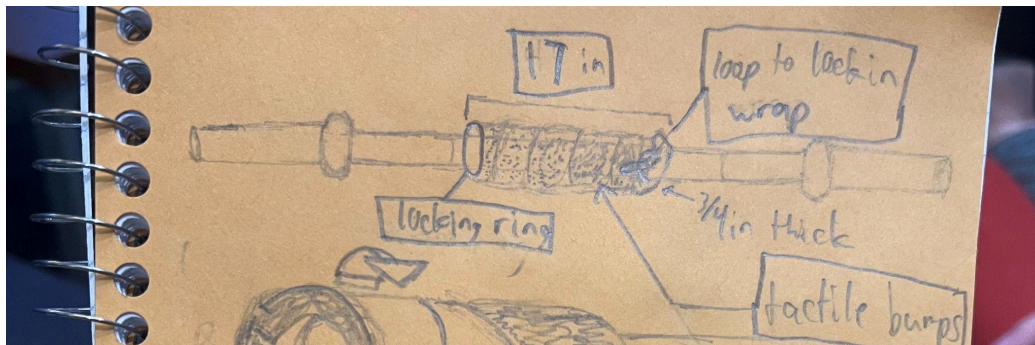
Product Breakdown

- Full-length wrap (~45") - covers the full contact area of a standard squat pads

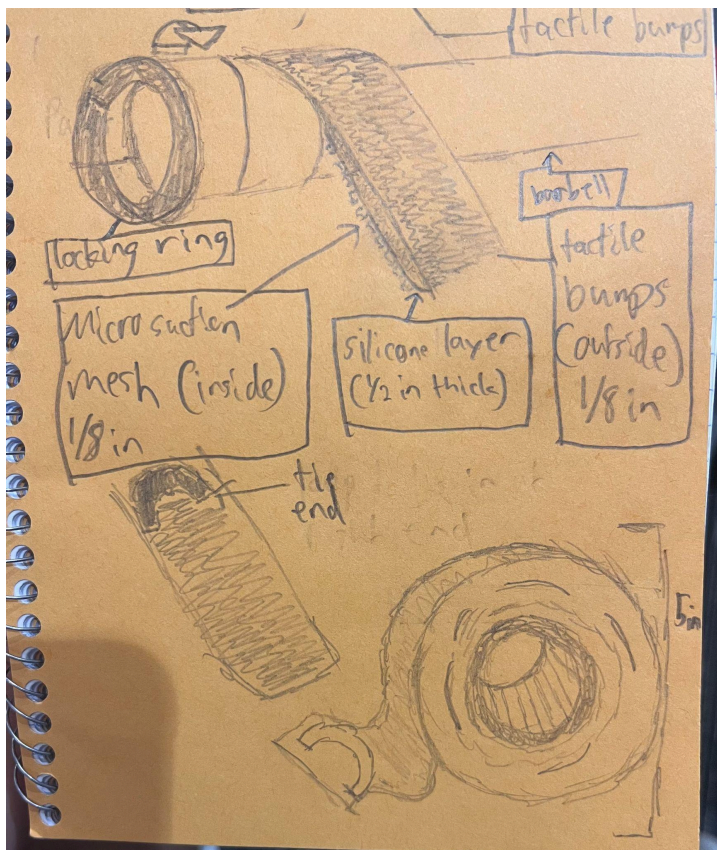
- Locking ring anchor (left) - prevents shifting or rolling mid-lift.
- Wrap-and-tie fastening (right) - adjustable tightness and easy release.
- Micro-suction mesh interior - creates a vacuum-seal grip to the bar.
- $\frac{1}{8}$ " tactile bumps - distribute pressure evenly while enhancing grip.
- Transparent minimalist design - discreet, hygienic, and professional look.

Illustrations

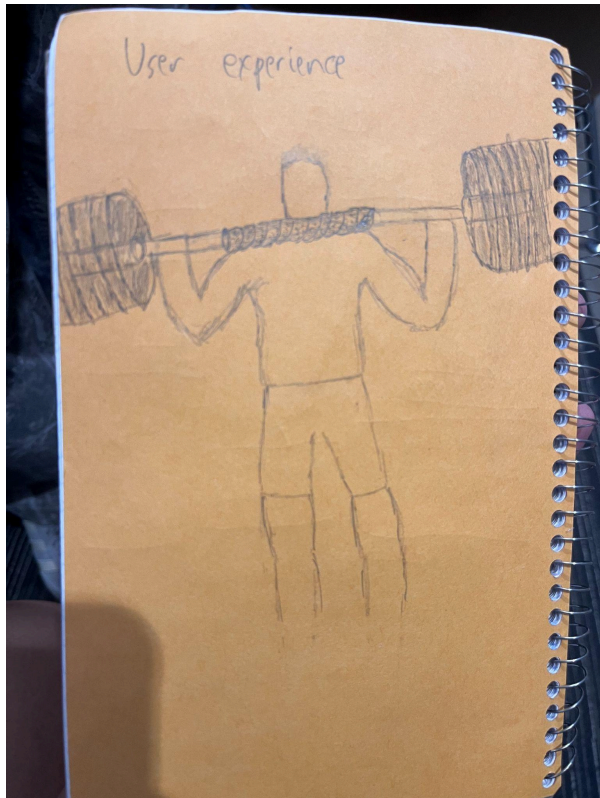
Whole



Parts



User Experience



Reflective Discussion

In developing this product we had great success in coming up with the initial idea. The idea of a discreet squat pad that has similar qualities of a normal protective squat pad, and was discreet was a great baseline solution. The main part that we struggled with however was the very detailed and business orientated aspects of developing the product or the “opportunity” portion of it. Identifying the details of our solution, and exactly how it would work was difficult. Also, identifying the gaps in the market was pretty tough for us, as we found it difficult to imagine indirect solutions. Really looking into the facts of who would be interested in this product or where would our project even start was also something that we struggled with.

Looking ahead, one challenge our team anticipates in the next phase is accurately forecasting and preparing for the pitch competition. We expect difficulties in predicting what specific financial, technical, or market questions the judges may ask, especially those that touch on niche details or assumptions we haven’t fully modeled yet. Ensuring our resource plan and launch strategy are both realistic and defensible will require careful attention to data and scenario planning. Despite that challenge, we are highly optimistic about our position because of how clearly our product differentiates itself from existing solutions. Our concept offers a unique value proposition that combines practicality, affordability, and user-centered innovation, something

both novel and feasible in real-world application of squat supporting technology. This differentiation gives us confidence that, if we communicate our story clearly and emphasize the tangible value our product provides, we can stand out among competitors and make a strong impression during the pitch.